

BZT2

Physics: Waves and Optics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

Items

1.

A marine biologist on a research vessel with a crew of 20 records a 500-Hz whale call traveling through seawater at 1,500 m/s. What is the call's wavelength in the water?

- A. 750,000 m
- B. 3.0 m
- C. 0.33 m
- D. About 0.69 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A sound wave passes from air into water, where it travels faster. What happens to its frequency and wavelength?

- A. Both increase.
- B. Frequency stays the same; wavelength increases.
- C. Neither changes, since waves never change when crossing a boundary.
- D. Frequency increases; wavelength stays the same.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a 3,000-seat concert venue, a technician measures 90 dB at the back of the hall and 110 dB next to a speaker. How many times more intense is the sound at the speaker?

- A. 100 times
- B. About 1.2 times
- C. 20 times
- D. 10 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An ambulance from a county service with 14 vehicles sounds a 700-Hz siren as it approaches a pedestrian, passes, and drives away. What does the pedestrian hear?

- A. 700 Hz the whole time.
- B. Below 700 Hz while it approaches, above 700 Hz as it moves away.
- C. Higher pitch whenever it is loudest, since louder sounds have higher pitch.
- D. Above 700 Hz while it approaches, below 700 Hz as it moves away.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
-----------------------------	---	----------------------	-------------------------------

5.

A guitar shop that repairs 50 instruments a month restrings a guitar whose string vibrates over 0.65 m between fixed ends. What is the wavelength of the fundamental?

- A. 0.325 m
- B. 0.65 m
- C. 2.60 m
- D. 1.30 m

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
-----------------------------	---	----------------------	-------------------------------

6.

Sunlight in air strikes a window in a 40-story office tower at 30° from the normal. The glass has $n = 1.50$. What is the angle of refraction inside the glass?

- A. About 48.6°
- B. 30° , since light never bends in solid glass.
- C. About 19.5°
- D. 45°

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
-----------------------------	---	----------------------	-------------------------------

7. [TRANSFER — reversed stem / different variable]

Light inside glass ($n = 1.50$) reaches the glass-air surface at 30° from the normal. What is the angle in the air?

- A. About 48.6°
- B. The light is totally reflected.
- C. About 19.5°
- D. 30°

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
-----------------------------	---	----------------------	-------------------------------

8.

A photographer who shoots weddings on weekends uses a converging lens with a 10-cm focal length. The subject is 30 cm from the lens. Where does the image form?

- A. 15 cm on the far side (real, inverted)
- B. 7.5 cm
- C. 20 cm
- D. 40 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A converging lens with $f = 10$ cm forms a real image 15 cm from the lens. How far is the object from the lens?

- A. 25 cm
- B. 30 cm
- C. 6 cm
- D. 5 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Gas in the cylinder of a pump designed by an engineering firm in Ohio absorbs 500 J of heat and does 200 J of work pushing a piston outward. What is the change in the gas's internal energy?

- A. -300 J
- B. +700 J
- C. +300 J
- D. +500 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A gas's internal energy rises by 150 J while 400 J of work is done on the gas. How much heat did the gas exchange?

- A. It absorbed 250 J of heat.
- B. It absorbed 550 J of heat.
- C. It absorbed 400 J of heat.
- D. It released 250 J of heat.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A tank of helium at a party-supply store with two locations is heated from 27 °C to 327 °C. By what factor does the average kinetic energy of the helium atoms change?

- A.** About 12
- B.** $\sqrt{2}$
- C.** 2
- D.** 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A marine biologist on a research vessel ~~with a crew of 20~~ records a 500-Hz whale call traveling through seawater at 1,500 m/s. What is the call's wavelength in the water?

Target word: wavelength **Decides:** $\lambda = v / f$.

A	Inverted operation	Multiplies $v \times f$.
B	Correct	$1,500 / 500 = 3.0$.
C	Confusion set	Divides f by v .
D	Adjacent category	Uses the speed of sound in air, 343 m/s.

2. Answer B Concept 1 • transfer

A sound wave passes from air into water, where it travels faster. What happens to its frequency and wavelength?

Target word: which variable adjusts **Decides:** The source fixes frequency.

A	Confusion set	Frequency cannot change at the boundary.
B	Correct	$v = f\lambda$ with f fixed, so λ grows with v .
C	Absolute language	Speed and wavelength do change.
D	Inverted operation	Swaps which quantity the medium changes.

3. Answer A Concept 2

At a ~~3,000-seat~~ concert venue, a technician measures 90 dB at the back of the hall and 110 dB next to a speaker. How many times more intense is the sound at the speaker?

Target word: intensity ratio **Decides:** 20 dB difference = 10^2 .

A	Correct	Two 10-dB steps, each $\times 10$.
B	Adjacent category	Divides decibel values as if the scale were linear.
C	Wrong stem number	Uses the dB difference as the ratio.
D	Dropped element	Counts only one 10-dB step.

4. Answer D Concept 3

An ambulance ~~from a county service with 14 vehicles~~ sounds a 700-Hz siren as it approaches a pedestrian, passes, and drives away. What does the pedestrian hear?

Target word: observed pitch **Decides:** Approaching raises, receding lowers the heard frequency.

A	Dropped element	Drops the relative motion.
B	Inverted operation	Reverses the shift.
C	Confusion set	Loudness (amplitude) is not pitch (frequency).
D	Correct	Wavefronts bunch up ahead and spread out behind.

5. Answer D Concept 4

A guitar shop that repairs 50 instruments a month restrings a guitar whose string vibrates over 0.65 m between fixed ends. What is the wavelength of the fundamental?

Target word: fundamental wavelength **Decides:** Fixed-fixed: $\lambda = 2L$.

A	Inverted operation	Halves L instead of doubling it.
B	Dropped element	Sets λ equal to L.
C	Adjacent category	Uses the closed-open pipe rule, $4L$.
D	Correct	2×0.65 .

6. Answer C Concept 5

Sunlight in air strikes a window in a 40-story office tower at 30° from the normal. The glass has $n = 1.50$. What is the angle of refraction inside the glass?

Target word: refracted angle **Decides:** $\sin \theta_2 = \sin 30^\circ / 1.50 = 0.333$.

A	Inverted operation	Multiplies by 1.50, bending away from the normal.
B	Absolute language	Light bends at any boundary between different indexes.
C	Correct	$\arcsin(0.333) \approx 19.5^\circ$, bent toward the normal.
D	Confusion set	Multiplies the angle, not its sine, by n.

7. Answer A Concept 5 • transfer

Light inside glass ($n = 1.50$) reaches the glass-air surface at 30° from the normal. What is the angle in the air?

Target word: reverse direction of travel **Decides:** $\sin \theta_{\text{air}} = 1.50 \times \sin 30^\circ = 0.75$.

A	Correct	$\arcsin(0.75) \approx 48.6^\circ$, bent away from the normal.
B	Confusion set	The critical angle is about 41.8° ; 30° is below it.
C	Inverted operation	Treats it as entering glass.
D	Dropped element	Drops the index change.

8. Answer A Concept 6

A photographer who shoots weddings on weekends uses a converging lens with a 10-cm focal length. The subject is 30 cm from the lens. Where does the image form?

Target word: image distance **Decides:** $1/d_i = 1/10 - 1/30$.

A	Correct	$1/d_i = 2/30$, so $d_i = 15$ cm.
B	Inverted operation	Adds $1/10 + 1/30$ instead of subtracting.
C	Adjacent category	Subtracts the distances instead of their reciprocals.
D	Confusion set	Adds the distances.

9. Answer B Concept 6 • transfer

A converging lens with $f = 10$ cm forms a real image 15 cm from the lens. How far is the object from the lens?

Target word: solve for object distance **Decides:** $1/d_o = 1/10 - 1/15$.

A	Confusion set	Adds distances.
B	Correct	$1/d_o = 1/30$.
C	Inverted operation	Adds the reciprocals.
D	Adjacent category	Subtracts distances.

10. Answer C Concept 7

Gas in the cylinder of a pump ~~designed by an engineering firm in Ohio~~ absorbs 500 J of heat and does 200 J of work pushing a piston outward. What is the change in the gas's internal energy?

Target word: ΔU **Decides:** $\Delta U = Q - W_{\text{by}} = 500 - 200$.

A	Confusion set	Reverses both signs.
B	Inverted operation	Adds work done by the gas.
C	Correct	Heat in minus work out.
D	Dropped element	Drops the work term.

11. Answer D Concept 7 • transfer

A gas's internal energy rises by 150 J while 400 J of work is done on the gas. How much heat did the gas exchange?

Target word: solve for Q **Decides:** Work on the gas is $W_{\text{by}} = -400$ J: $Q = \Delta U + W_{\text{by}} = 150 - 400$.

A	Confusion set	Right size, wrong direction.
B	Inverted operation	Treats work on the gas as work by it.
C	Wrong stem number	Uses the work figure as the heat.
D	Correct	$Q = -250$ J.

12. Answer C Concept 8

A tank of helium ~~at a party supply store with two locations~~ is heated from 27 °C to 327 °C. By what factor does the average kinetic energy of the helium atoms change?

Target word: KE factor **Decides:** 300 K $\rightarrow$ 600 K.

A	Dropped element	Uses Celsius: 327/27.
B	Adjacent category	That is the rms speed factor, not energy.
C	Correct	$KE \propto T$ in kelvin; $600/300 = 2$.
D	Confusion set	Squares the temperature ratio.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Wave speed, frequency, wavelength	$v = f\lambda$; the source sets frequency, the medium sets speed, so wavelength adjusts when the medium changes.	Thinking frequency changes when a wave enters a new medium.	A wave crossing a boundary, or two of v , f , λ given.
2	Decibel scale	Every +10 dB multiplies intensity by 10; +20 dB is 100 $\times$.	Treating decibels as a linear scale.	Two sound levels in dB compared.
3	Doppler effect	A source moving toward a listener is heard at a higher frequency; moving away, lower.	Confusing loudness with pitch.	A siren or horn passing a listener.
4	Standing waves	A string fixed at both ends (or an open-open pipe) has a fundamental wavelength of $2L$; a closed-open pipe has $4L$.	Setting the fundamental wavelength equal to the length.	A string or pipe length and “fundamental.”
5	Refraction and Snell's law	$n_1 \sin \theta_1 = n_2 \sin \theta_2$; light bends toward the normal entering a higher-index medium and away from it leaving.	Multiplying or dividing the angle itself, or bending the wrong way.	Two indexes and an angle of incidence.
6	Thin lens equation	$1/f = 1/d_o + 1/d_i$; for a converging lens with the object beyond f , the image is real and inverted.	Adding reciprocals where one must be subtracted, or adding distances.	A focal length and one of the two distances.
7	First law of thermodynamics	$\Delta U = Q - W$, where Q is heat added to the gas and W is work done by the gas.	Adding work done by the gas, or mixing sign conventions.	Heat and work amounts with “by” or “on” the gas.
8	Kinetic theory of gases	Average kinetic energy of gas particles is proportional to absolute temperature; rms speed goes as $\sqrt{T}$.	Using Celsius ratios, or confusing speed with energy.	A gas heated between two Celsius temperatures.

Course scope (public description): Physics: Waves and Optics addresses foundational topics in the physics of waves and optics. Students will study basic wave motion and then apply that knowledge to the study of sound and light with even further applications to optical instruments. This course will also cover thermodynamics and theories governing the physics of gases.